# SQL JOINS

```
mysql> CREATE DATABASE school_db;
Query OK, 1 row affected (0.01 sec)

mysql> USE school_db;
Database changed
mysql> CREATE TABLE Student (
    -> student_id INT PRIMARY KEY,
    -> Student_name VARCHAR(100) NOT NULL,
    -> phone_no INT,
    -> depatment_id INT
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> INSERT INTO Student (student_id, Student_name, phone_no, depatment_id) VALUES
    -> (12, 'Aarav Sharma', 987654321, 1),
    -> (29, 'Diya Patel', 876543210, 2),
    -> (19, 'Vivaan Reddy', 765432109, 3),
    -> (34, 'Ananya Iyer', 654321098, 1),
    -> (54, 'Kabir Singh', 543210987, 4),
    -> (13, 'Ruthu Gowda', 783990987, 2);
Query OK, 6 rows affected (0.02 sec)
Records: 6  Duplicates: 0  Warnings: 0
```

```
mysql> CREATE TABLE department1 (
    ->  department_id INT,
    -> department_name VARCHAR(100) NOT NULL
    -> );
Query OK, 0 rows affected (0.03 sec)
```

```
ysql> INSERT INTO department1 (department_id, department_name) VALUES
    -> (1, 'Computer Science'),
    -> (2, 'Electronics Engineering'),
    -> (3, 'Mechanical Engineering'),
    -> (4, 'Information Technology');
uery OK, 4 rows affected (0.01 sec)
ecords: 4  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT Student.student_id, Student.Student_name, department1.department_name
    -> FROM Student
    -> LEFT JOIN department1 ON Student.depatment_id = department1.department_id;
+------------+--------------+------------------------+
| student_id | Student_name | department_name        |
+------------+--------------+------------------------+
|         12 | Aarav Sharma | Computer Science       |
|         13 | Ruthu Gowda  | Electronics Engineering |
|         19 | Vivaan Reddy | Mechanical Engineering  |
|         29 | Diya Patel   | Electronics Engineering |
|         34 | Ananya Iyer  | Computer Science       |
|         54 | Kabir Singh  | Information Technology  |
+------------+--------------+------------------------+
6 rows in set (0.00 sec)
```

```
mysql> SELECT Student.student_id, Student.Student_name, department1.department_name
    -> FROM Student
    -> INNER JOIN department1 ON Student.depatment_id = department1.department_id;
+------------+--------------+------------------------+
| student_id | Student_name | department_name        |
+------------+--------------+------------------------+
|         12 | Aarav Sharma | Computer Science       |
|         13 | Ruthu Gowda  | Electronics Engineering |
|         19 | Vivaan Reddy | Mechanical Engineering  |
|         29 | Diya Patel   | Electronics Engineering |
|         34 | Ananya Iyer  | Computer Science       |
|         54 | Kabir Singh  | Information Technology  |
+------------+--------------+------------------------+
6 rows in set (0.01 sec)
```

```
mysql> SELECT Student.student_id, Student.Student_name, department1.department_name
    -> FROM Student
    -> RIGHT JOIN department1 ON Student.depatment_id = department1.department_id;
+------------+--------------+------------------------+
| student_id | Student_name | department_name        |
+------------+--------------+------------------------+
|         34 | Ananya Iyer  | Computer Science       |
|         12 | Aarav Sharma | Computer Science       |
|         29 | Diya Patel   | Electronics Engineering |
|         13 | Ruthu Gowda  | Electronics Engineering |
|         19 | Vivaan Reddy | Mechanical Engineering  |
|         54 | Kabir Singh  | Information Technology  |
+------------+--------------+------------------------+
6 rows in set (0.00 sec)
```

```
mysql> SELECT Student.student_id, Student.Student_name, department1.department_name
    -> FROM Student
    -> LEFT JOIN department1 ON Student.depatment_id = department1.department_id
    -> UNION
    -> SELECT Student.student_id, Student.Student_name, department1.department_name
    -> FROM Student
    -> RIGHT JOIN department1 ON Student.depatment_id = department1.department_id;
+------------+--------------+------------------------+
| student_id | Student_name | department_name        |
+------------+--------------+------------------------+
|         12 | Aarav Sharma | Computer Science       |
|         13 | Ruthu Gowda  | Electronics Engineering |
|         19 | Vivaan Reddy | Mechanical Engineering  |
|         29 | Diya Patel   | Electronics Engineering |
|         34 | Ananya Iyer  | Computer Science       |
|         54 | Kabir Singh  | Information Technology  |
+------------+--------------+------------------------+
6 rows in set (0.01 sec)
```